$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\left(\hat{H}'_{\mathcal{S}}\hat{\rho}_{\mathcal{S}}(t) - \rho_{\mathcal{S}}(t)\hat{H}'^{\dagger}_{\mathcal{S}}\right) - \widetilde{D}\left[\hat{\sigma}_z, [\hat{\sigma}_z, \hat{\rho}_{\mathcal{S}}(t)]\right] + \zeta\hat{\sigma}_z\hat{\rho}_{\mathcal{S}}(t)\hat{\sigma}_y + \zeta^*\hat{\sigma}_y\hat{\rho}_{\mathcal{S}}(t)\hat{\sigma}_z. \quad (5.142)$$